

Investigation of Nanographene as Negative Electrode for Sodium-Ion Battery Storage Performance Using Machine Learning Potential and Grand Canonical Monte Carlo Method

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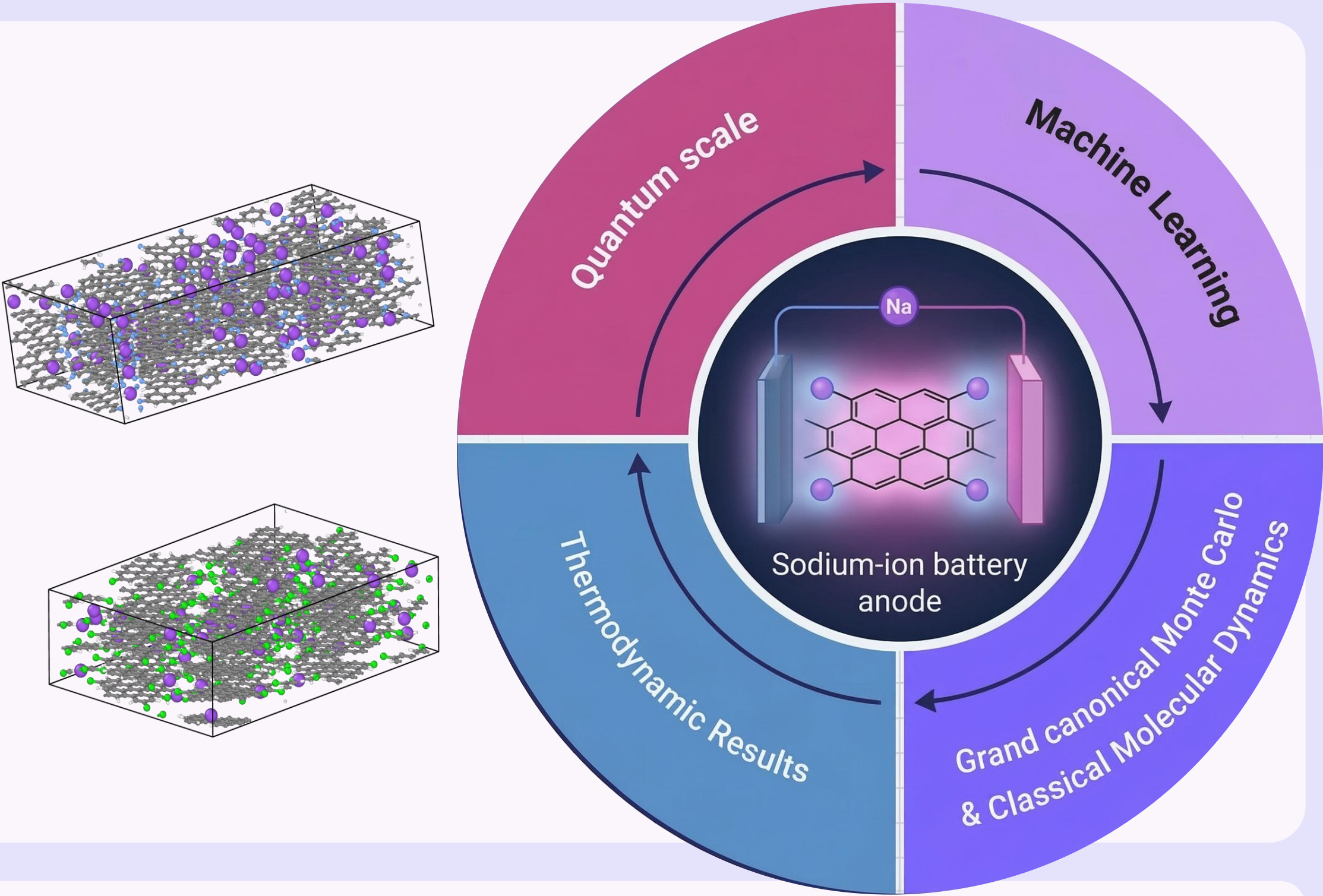
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Abstract

- **Sodium-ion batteries (SIBs)** are promising for large-scale energy storage, but the larger Na^+ radius causes sluggish intercalation in traditional graphite. To overcome this, **halogen-functionalized nanographenes (6F-HBC and 6Cl-HBC)** offer highly electronegative edges and structural flexibility, making them ideal models for exploring novel storage mechanisms.
- However, understanding the charge transfer between anode and sodium atom & the atomic-level volume expansion during high sodium loading remains a challenge. Experimental *in-situ* tools struggle to capture localized dynamic rearrangements, while traditional theoretical methods cannot simultaneously balance **quantum accuracy, large-scale systems, and open-system thermodynamics**.
- To bridge this gap, we propose a novel multi-scale framework: combining **DFT datasets, DeePMD machine learning potentials, and Hybrid GCMC/MD simulations**. This approach accurately captures chemical-potential-driven Na insertion and the continuous "structural breathing" effect of 6F/6Cl-HBC.



Methodology

- To accurately capture the complex sodium storage mechanisms—including both edge adsorption and interlayer intercalation—this study employs a multi-scale computational framework starting with **Density Functional Theory (DFT)**. We constructed a **four-molecule representative unit** of 6F-HBC and 6Cl-HBC to fully account for interlayer and lateral van der Waals interactions. A comprehensive dataset generated via **VASP** was enriched with high-energy repulsive configurations, nanographene interlayer distance scan, and sodium cluster interactions to ensure **physical robustness** during macroscopic sampling.
- Building upon the DFT data, we trained a highly efficient machine learning potential using the **DeePMD-kit** with the **se_e2_a** descriptor to achieve quantum-level accuracy.
- Finally, the structural evolution and theoretical capacity were evaluated using Hybrid GCMC/MD simulations in **LAMMPS**, utilizing a large-scale supercell **containing 32 nanographene molecules** (1,920 atoms). While GCMC manages the chemical-potential-driven insertion and deletion of sodium ions, intermittent **NPT-MD** steps allow the carbon framework to continuously relax and expand. This quasi-static coupling perfectly captures the dynamic "structural breathing" effect and reveals the true thermodynamic storage pathways at high sodium concentrations.

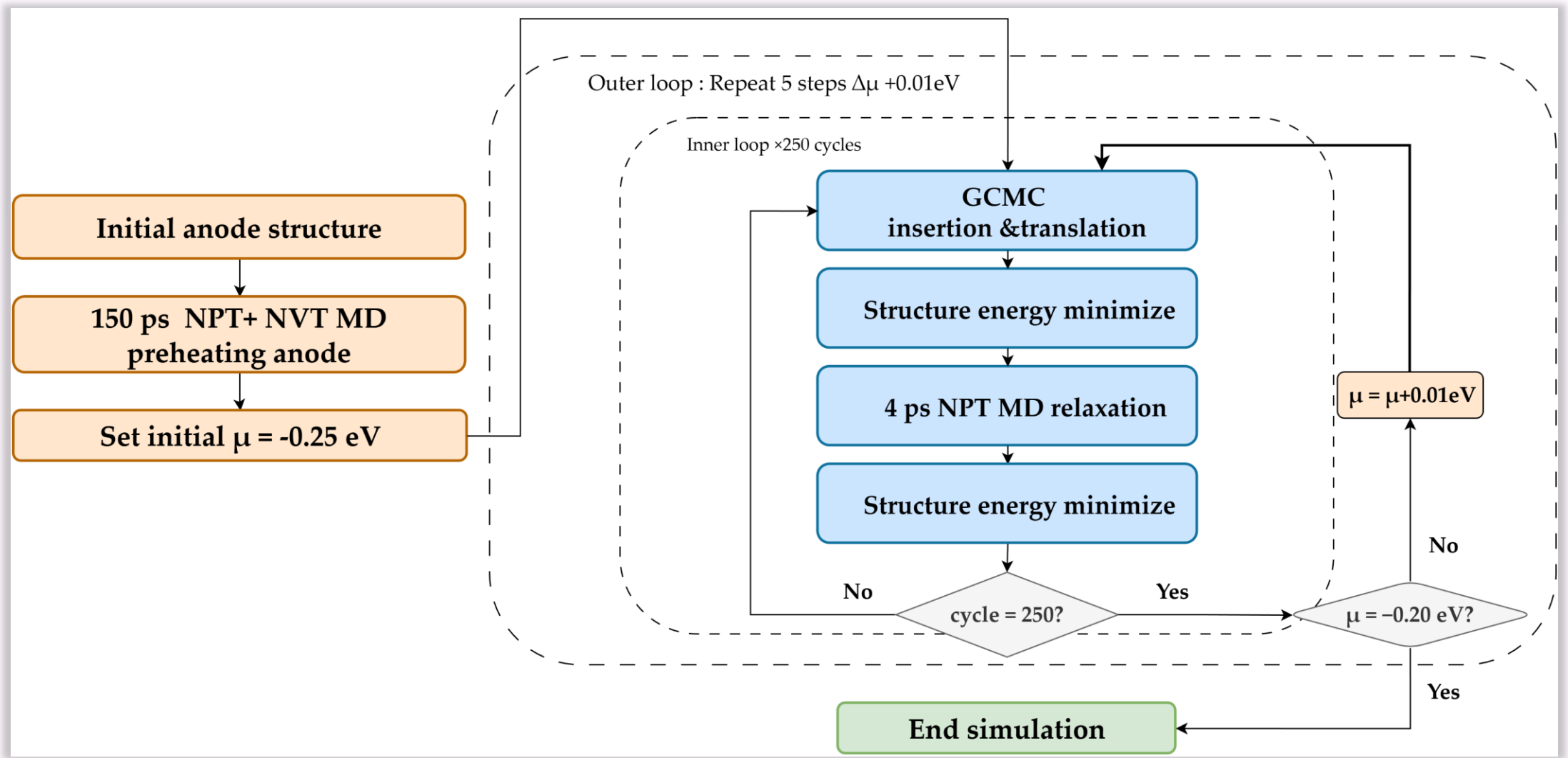


Fig. 1 : Schematic Workflow of Hybrid GCMC & NPT MD simulation

Results

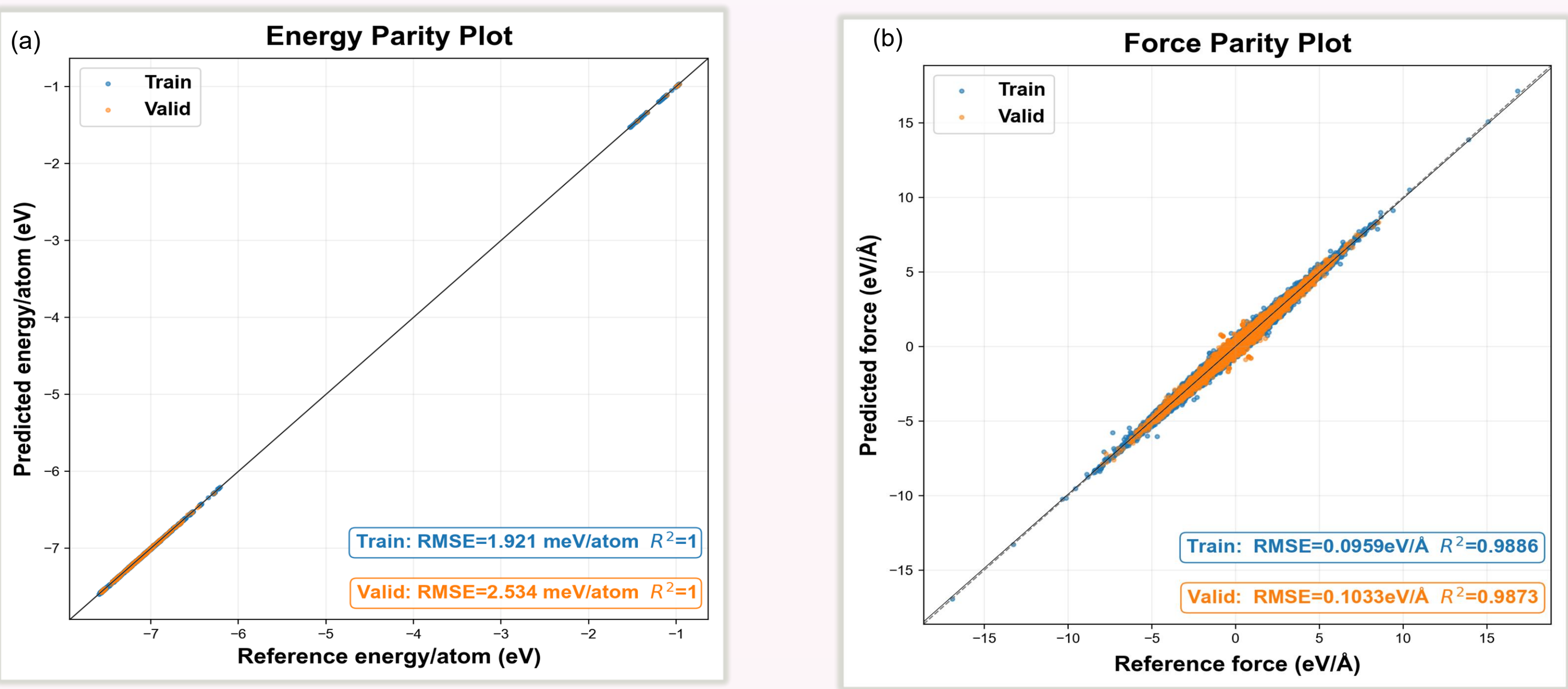


Fig. 2 : Machine learning force field highly reproduce DFT result (a) energy parity plot (b) force parity plot

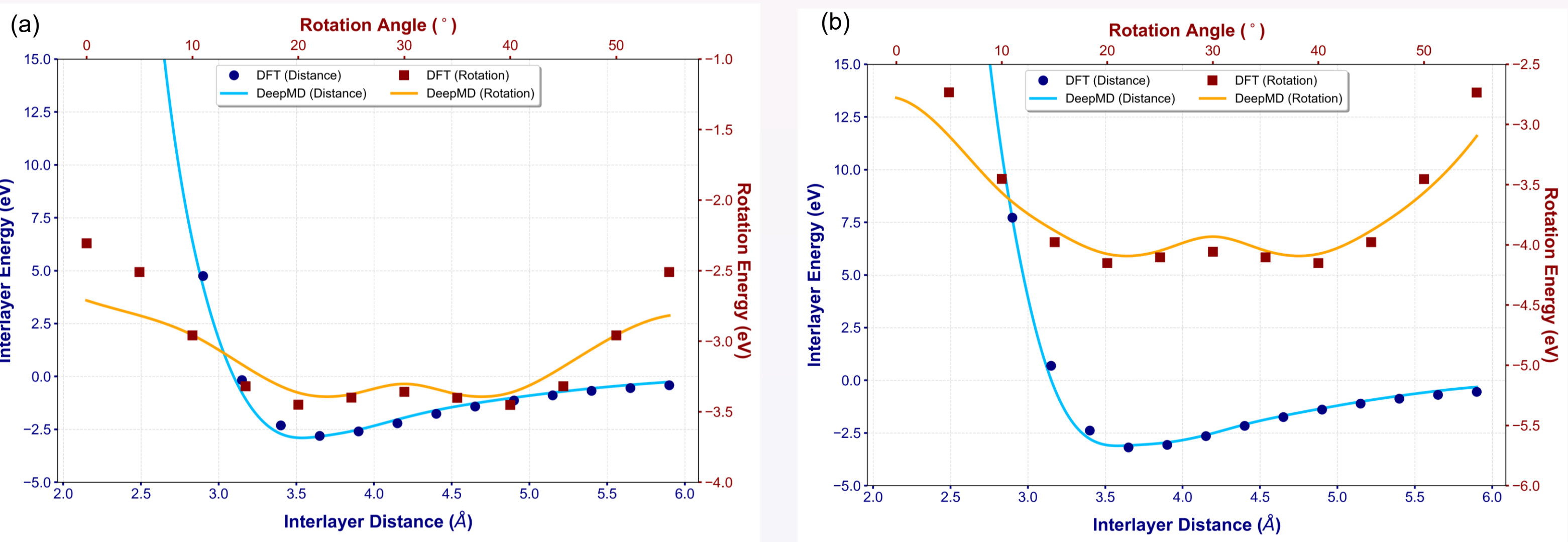


Fig. 3 : Interlayer interaction PES of (a) 6F-HBC (b) 6Cl-HBC

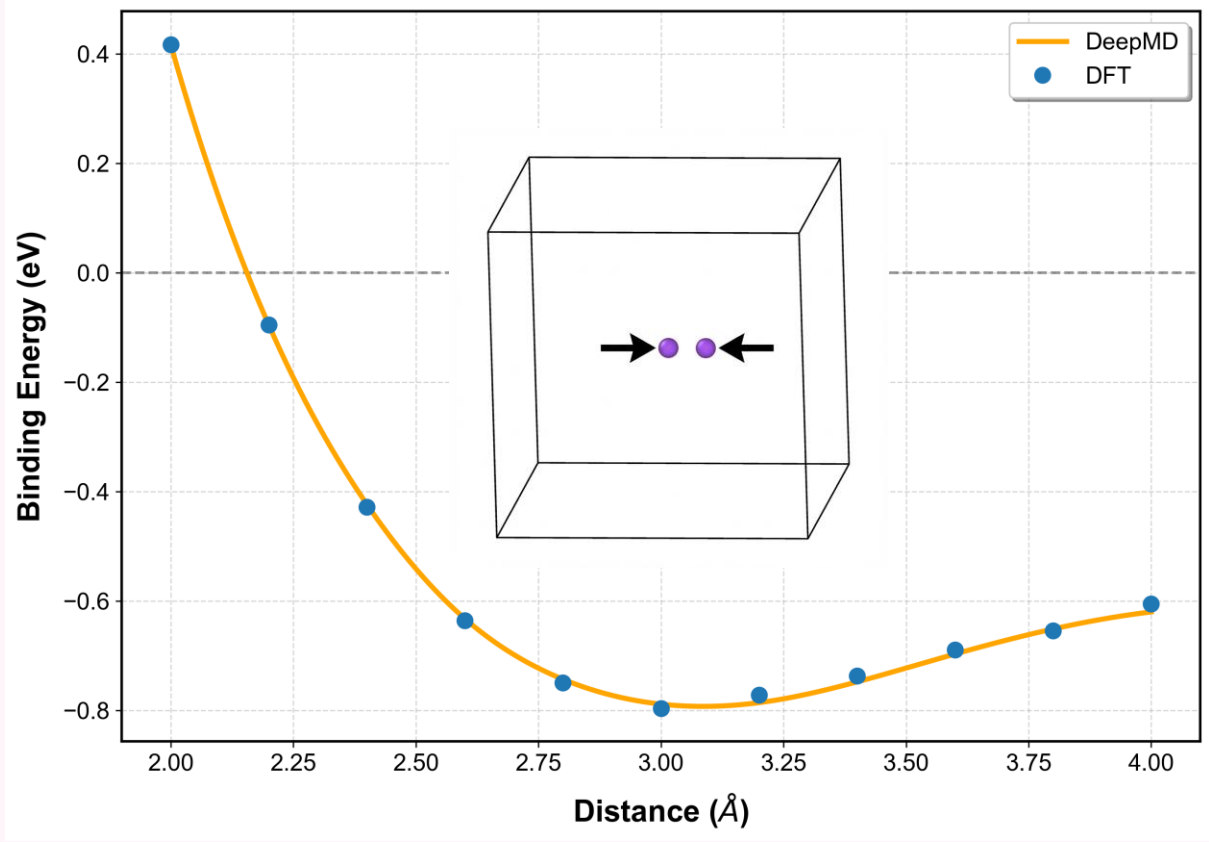


Fig. 4 : Sodium dimer Binding energy PES curve in good agreement with DFT calculation

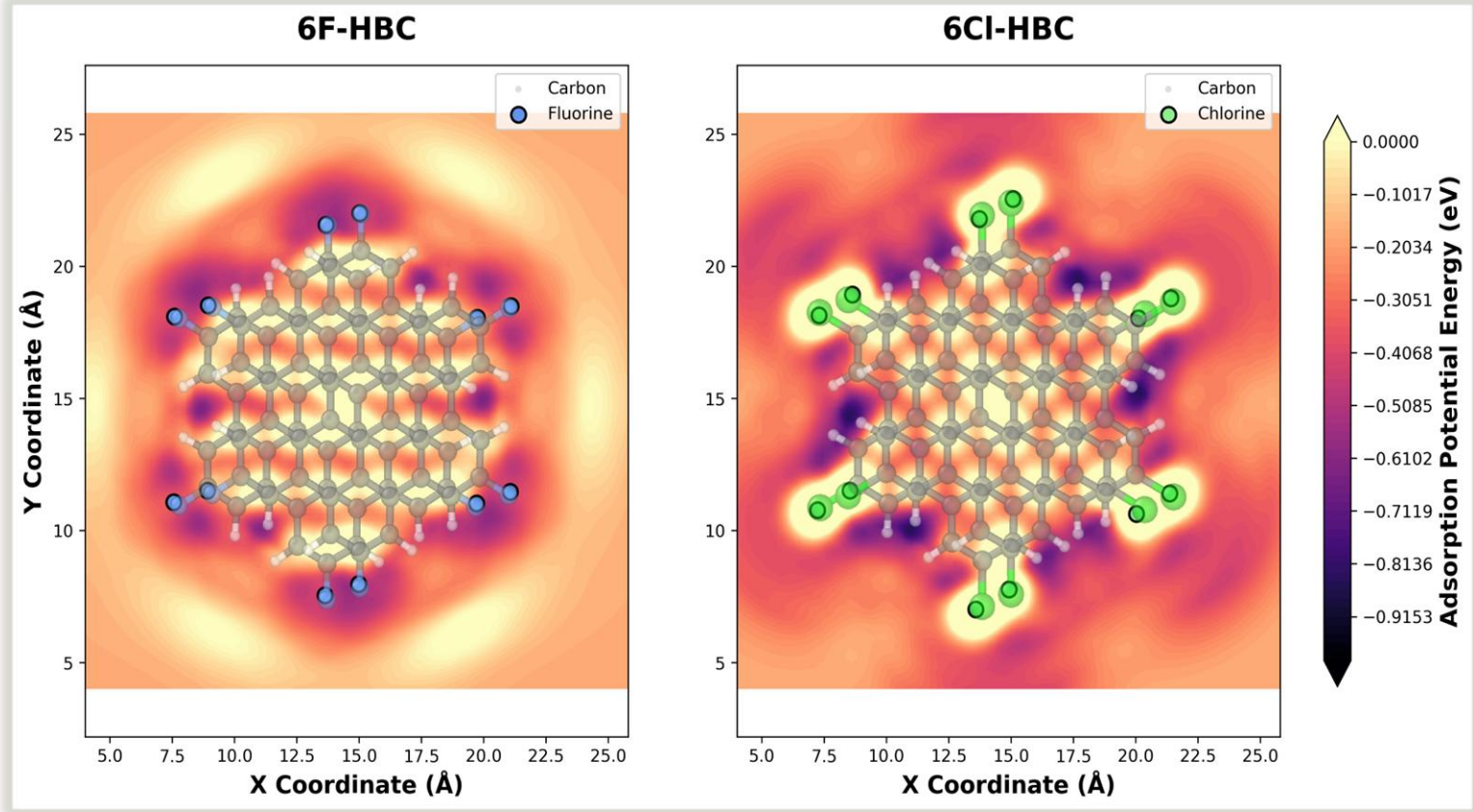


Fig. 5 : 2D Adsorption Energy Surface Scan Calculation shows that sodium atom prefer stay at the bay area surrounding by halogen and hydrogen

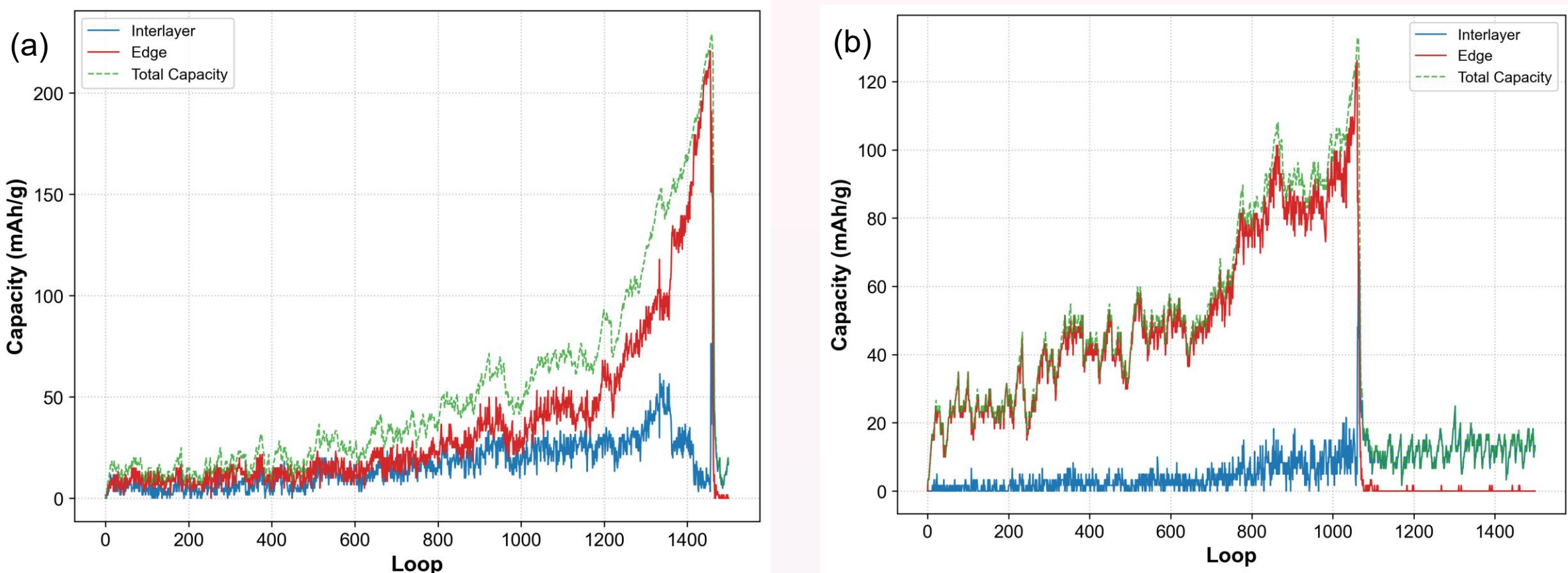


Fig. 6 : these curve identify the capacity contribution of which location (a)6F-HBC (b)6Cl-HBC

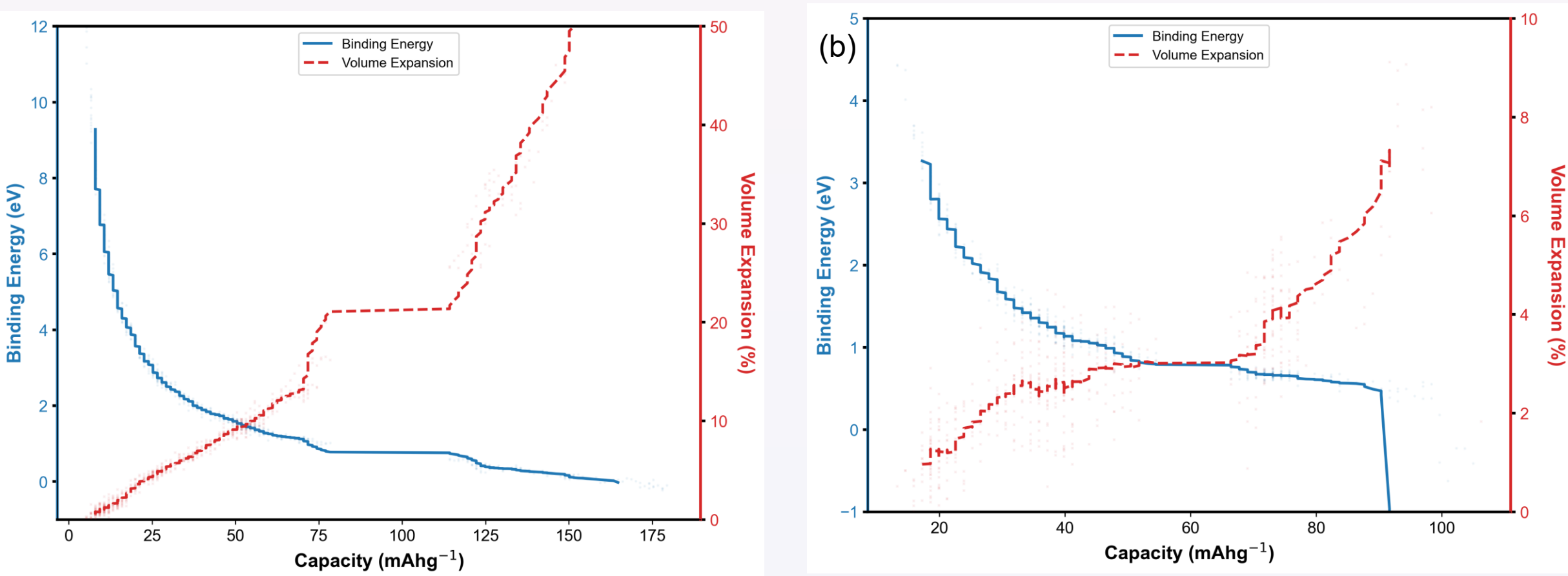
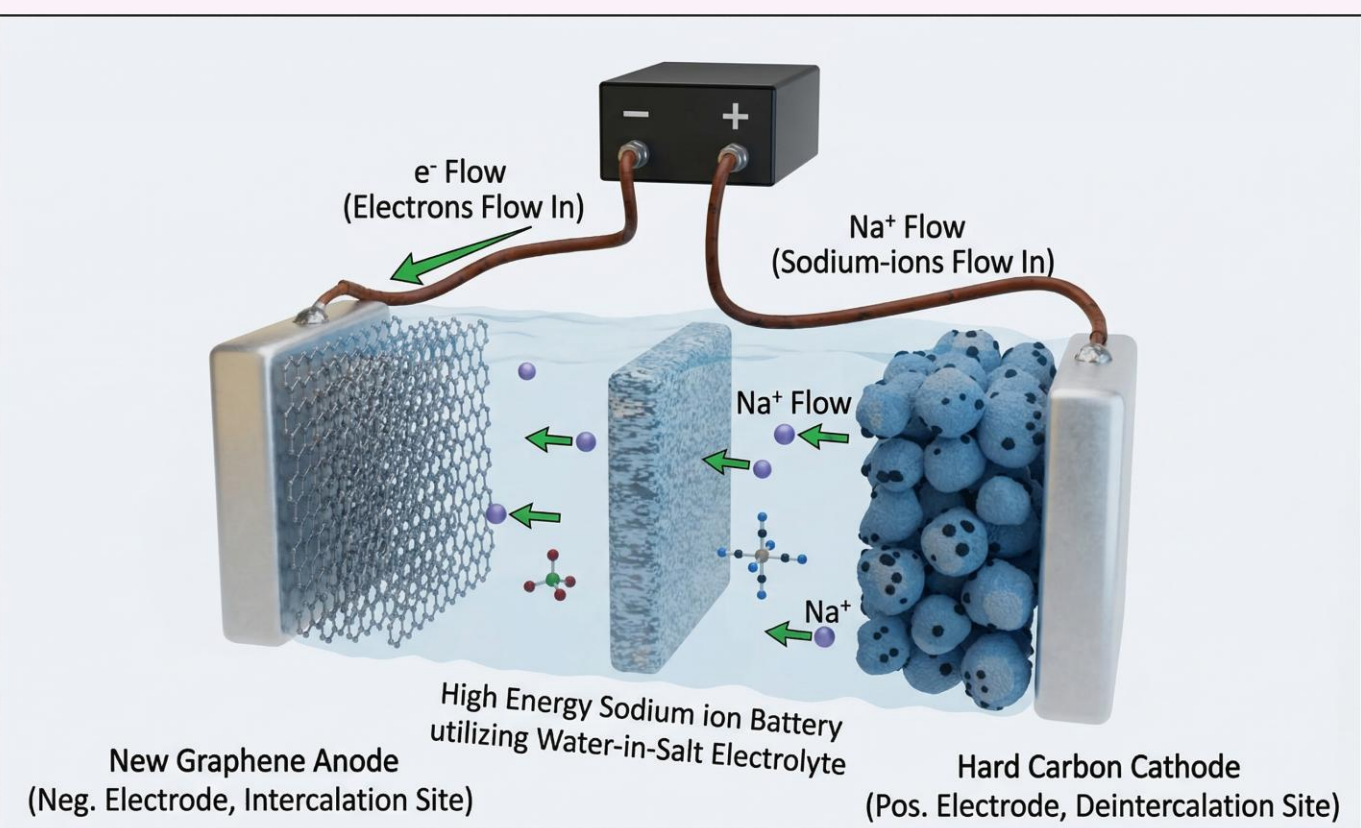


Fig. 7 : Binding energy & volume expansion rate vs. Capacity plot shows the limitation capacity of two anode structure, 6F-HBC with extremely large volume expansion rate, 6Cl-HBC with poor capacity (a) 6F-HBC (b) 6Cl-HBC

Conclusion

- GCMC simulation results reveal that while nanographene exhibits high flexibility and edge adsorption, its capacity remains limited due to a **lack of intercalation**
- Machine learning potentials successfully extend quantum-level accuracy to classical molecular dynamics (MD), enabling simulations at much larger spatial and temporal scales.
- Integrated with Grand Canonical Monte Carlo method allows us to investigate the configuration space of anode structures and identify thermodynamically preferred site locations and insertion sequences



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